DipTrace For Your SCHEMATIC AND PCB DESIGN

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The electronics industry is expanding like never before. But every application is dependent on that tiny block called printed circuit board (PCB), which is totally hidden by wires and connected devices. The next time you start a project, why not design your own PCB, instead of getting someone else to do it for you? Here is how you can do it with DipTrace.

DipTrace is a PCB designing software that helps you design a neat and tidy board that works as per your needs. But when so many such software are available why should you choose to go with this specific tool? Let us try to find an answer to this question by looking at what the software offers for every stage of designing.

Easier sub-section design. While designing a circuit, it is always comfortable to make small parts that you know would work for sure and then piece these together to form the complete circuit. But this is not as easy as it seems, as there are complications with making sure the signals flow correctly and with ensuring common ground connections.

You will find this problem easier to handle in DipTrace, where each subsection can be designed on a separate sheet. The first sheet becomes the face of your design, to which the others can be linked hierarchically using connectors between sheets.

For nodes like power supplies or repeated ones in the design, you simply have to connect the nets to the same net port. The nets are then automatically linked across sheets, without wires. You can simply use a bus connection for groups of wires that go together. Nets on different sheets are connected by names, while buses on different sheets have exclusive connectors.

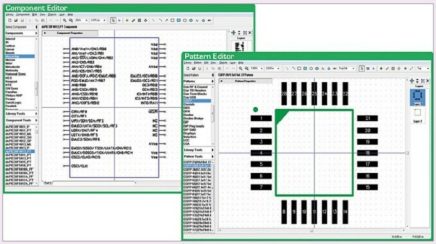


Fig. 1: Make your PCB from scratch

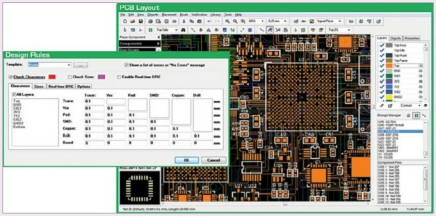


Fig. 2: Convert your schematic to a PCB layout

Once the schematic is complete, DipTrace allows you to cross-check parameters you might have missed, like shorts, or to super-impose pins by an electrical rule check (ERC) that is flexible in allowing you to choose the features you want to verify.

Pick and place your layout. Moving from the schematic design to the PCB layout stage, you can take forward the same schematic rules or import design rules from an earlier project. On importing non-schematic rules, your PCB is ready with net classes, layers and design constraints.

One of the major issues in creating layouts is the size: whether it will fit within the board it has to be printed on. This process is smooth